Nonlinear Convolution Equations, Spectral Rigidity, and the Sharp Restriction Constant on S^1

A Compactness Route to Finiteness of Critical Values

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1 Overview and Strategy

The ultimate target of this programme is the sharp Fourier extension constant on the unit circle $S^1 \subset \mathbb{R}^2$:

$$\mathcal{R} := \sup_{0 \neq f \in L^2(S^1)} \frac{\left\| \widehat{f\sigma} \right\|_{L^6(\mathbb{R}^2)}}{\|f\|_{L^2(S^1)}},$$

where σ is arclength measure on S^1. Every extremiser satisfies a nonlinear convolution equation on the sphere that fits into a general framework studied in the first part of this note.

1.1 The main strategy

After normalising $\|f\|_{L^2} = 1$ and writing $u = \widehat{f\sigma}$, an extremiser satisfies the Euler–Lagrange equation

$$u = c\big((|u|^4 u) * \widehat{\sigma}\big), \qquad c = \Lambda^{-1}, \qquad \Lambda = \|u\|_{L^6}^6 .$$

The strategy is:

1. **Spectral support forcing.** Show that $\widehat{u}$ is supported on S^1, reducing from all of $\mathbb{R}^2$ to a density $\rho \in L^2(S^1)$. **[Proved]**

2. **Fredholm structure.** Prove that the linearisation of the Euler–Lagrange map at any critical point is Fredholm of index zero, so that solutions are locally isolated modulo symmetry. **[Proved]**

3. **Compactness modulo symmetry.** Show that the set of normalised critical points is sequentially compact after quotienting by the symmetry group $G = U(1) \times SO(2) \ltimes \mathbb{R}^2$. **[Open]**

4. **Nondegeneracy.** At each critical point, verify that the kernel of the linearisation equals the tangent space to the symmetry orbit. At the constant solution, this reduces to finitely many Bessel integral inequalities. **[Conditional]**

5. **Finiteness and identification.** Conclude that the critical values Λ_j are finitely many, and identify the sharp constant as $\mathcal{R} = \max_j \Lambda_j^{1/6}$. **[Conditional]**

1.2 Status summary

Component	Status	Key input
Spectral support forcing (any n)	**[Proved]**	Paley–Wiener + support calculus
Compactness of T on PW (general)	**[Proved]**	Rellich for PW spaces
Compactness of K_{u_0} on $L^2(S^1)$	**[Proved]**	Truncation + Hilbert–Schmidt
Fredholm structure ($I - K$ index 0)	**[Proved]**	Riesz–Schauder
1D case (S^0)	**[Proved]**	Explicit algebraic solution
Nondegeneracy at constant solution	**[Conditional]**	Bessel integrals $\mathcal{J}_n$
Existence of maximiser	**[Proved]**	Profile decomposition + strict convexity
Compactness modulo G (all critical pts)	**[Open]**	Ruling out multi-profile splitting
Finiteness of critical values	**[Conditional]**	Items 3 + 4 above
Higher-dimensional generalisations	**[Open]**	Incidence-fibre smoothing

Part I
General Theory

2 Setup and Notation

We work throughout with *entire functions of finite exponential type*. An entire function $u : \mathbb{C} \to \mathbb{C}$ has *exponential type* $\tau \geq 0$ if $|u(z)| \leq Ce^{\tau|z|}$ for some $C > 0$ and all $z \in \mathbb{C}$. The *Paley–Wiener–Schwartz theorem* identifies this class: u is entire of type $\leq \tau$ with $u|_{\mathbb{R}} \in L^2(\mathbb{R})$ if and only if $\widehat{u} \in L^2$ is supported in $[-\tau, \tau]$. We write PW_τ for the corresponding Hilbert space.

Fourier conventions:

$$\widehat{u}(\xi) = \int_{\mathbb{R}} u(x)e^{-ix\xi}\,dx, \qquad u(x) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{u}(\xi)e^{ix\xi}\,d\xi,$$

so $\widehat{u * v} = \widehat{u} \cdot \widehat{v}$ and $\widehat{u \cdot v} = \frac{1}{2\pi}\widehat{u} * \widehat{v}$.

Throughout, $k \geq 1$ is a fixed positive integer, $c \in \mathbb{C} \setminus \{0\}$ is a fixed constant, and f is an entire function of exponential type $\tau_f > 0$ whose restriction to $\mathbb{R}$ lies in $L^2(\mathbb{R})$.

3 The Two Convolution Equations

3.1 The equation $u^k = c\,(u * f)$: spectral expansion kills solutions

In Fourier space the equation reads $\widehat{u}^{*k}(\xi) = c\,\widehat{u}(\xi)\widehat{f}(\xi)$.

Proposition 3.1 (Type obstruction)**.** *Let $u \in \mathrm{PW}_\tau$ with $\tau > 0$ satisfy $u^k = c(u * f)$ for $k \geq 2$. Then $u \equiv 0$.*

Proof. The Titchmarsh convolution theorem gives $\mathrm{supp}(\widehat{u}^{*k}) \subseteq [-k\tau, k\tau]$, while the right side is supported in $[-\tau, \tau]$. For $k \geq 2$ this forces $\tau = 0$, so u is a polynomial; degree comparison then forces u to be constant, and the only constants satisfying the equation are the $(k-1)$-th roots of $c\widehat{f}(0)$ together with 0. □

Corollary 3.2. *For $k \geq 2$ the solution set of $u^k = c(u * f)$ in the Paley–Wiener class is a finite discrete set of at most k constants.*

3.2 The equation $u = c\,(u^k * f)$: compact fixed-point structure

We now turn to the equation

$$u = c\,(u^k * f). \tag{1}$$

In Fourier space:

$$\widehat{u}(\xi) = c\,\widehat{u}^{*k}(\xi) \cdot \widehat{f}(\xi). \tag{2}$$

Proposition 3.3 (Type self-consistency)**.** *Any solution $u \in \mathrm{PW}_\tau$ of (1) satisfies $\tau \leq \tau_f$.*

Proof. The right side of (2) is supported in $\mathrm{supp}(\widehat{u}^{*k}) \cap \mathrm{supp}(\widehat{f}) \subseteq [-\tau_f, \tau_f]$. □

This is the key contrast: the equation $u = c(u^k * f)$ does *not* force $\tau = 0$; solutions of positive exponential type exist as long as $\tau \leq \tau_f$.

Proposition 3.4 (Compactness of T on Paley–Wiener spaces)**.** *Define $T : \mathrm{PW}_{\tau_f} \to \mathrm{PW}_{\tau_f}$ by $T(v) = c(v^k * f)$. Then T maps bounded sets in PW_τ (for any $\tau > \tau_f$) to precompact sets in PW_τ. In particular, $T : \mathrm{PW}_\tau \to \mathrm{PW}_\tau$ is a compact operator.*

Proof. Since $v \in \mathrm{PW}_{\tau_f}$ implies $v^k \in \mathrm{PW}_{k\tau_f}$, the function $v^k * f$ has Fourier support in $\mathrm{supp}(\widehat{v^k}) \cap \mathrm{supp}(\widehat{f}) \subseteq [-\tau_f, \tau_f]$. Hence T maps PW_τ into PW_{τ_f}, and the inclusion $\mathrm{PW}_{\tau_f} \hookrightarrow \mathrm{PW}_\tau$ is compact by the Rellich-type theorem for Paley–Wiener spaces. □

3.3 Fredholm structure and local manifold theorem

At a solution u_0 of (1), the Fréchet derivative of $G(u) := u - c(u^k * f)$ is

$$DG(u_0)[h] = h - ck(u_0^{k-1}h * f) =: (I - K)[h],$$

where $K : h \mapsto ck(u_0^{k-1}h * f)$ is compact by the same argument. By Riesz–Schauder, $I - K$ is **Fredholm of index zero**.

Theorem 3.5 (Local manifold structure)**.** *Let u_0 be a solution of* (1) *at which $DG(u_0) = I - K$ is invertible. Then u_0 is an isolated solution. More generally, the solution set near u_0 is a smooth manifold of dimension* $\dim\ker(I - K)$*, which is finite.*

Remark 3.6 (Direct contraction argument). For u, v in a ball of radius R in PW_{τ_f}, Young's inequality gives $\|T(u) - T(v)\| \leq |c|\, kR^{k-1} \|f\|_{L^1} \|u - v\|$. This is a contraction whenever $R < (|c|\, k \|f\|_{L^1})^{-1/(k-1)}$, giving local uniqueness without spectral theory.

3.4 Comparison of the two equations

	$u^k = c(u * f)$	$u = c(u^k * f)$
Fourier identity	$\widehat{u}^{*k} = c\widehat{u}\widehat{f}$	$\widehat{u} = c\widehat{u}^{*k} \cdot \widehat{f}$
Type constraint ($k \geq 2$)	Forces $\tau = 0$ (Titchmarsh)	Allows $\tau \leq \tau_f$
Operator structure	No compactness	T compact
Linearisation	Not Fredholm in general	$I - K$, Fredholm index 0
Solutions ($k \geq 2$, generic)	Finitely many constants	Isolated; Leray–Schauder applies

Table 1: Analytic contrast between the two convolution equations.

4 Spectral Support Forcing

We specialise to $f = \widehat{\sigma}$, where σ is the surface measure on $S^{n-1} \subset \mathbb{R}^n$. By the Paley–Wiener theorem, f is entire of exponential type 1, and $\widehat{f} = (2\pi)^n\sigma$ as a distribution.

Proposition 4.1 (Spectral support forcing)**.** *If u is entire of exponential type satisfying $u = c(u^k * \widehat{\sigma})$, then $\widehat{u}$ is a distribution supported on S^{n-1}. The same conclusion holds for the equation $u = c(|u|^{2(k-1)}u * \widehat{\sigma})$.*

Proof. From (2), $\widehat{u} = c\,\widehat{u}^{*k} \cdot (2\pi)^n \sigma$. The right side is supported on $\operatorname{supp}\sigma = S^{n-1}$.

For the $|u|^{2(k-1)}u$ variant: writing $|u|^{2(k-1)}u = u^k \bar{u}^{k-1}$ in Fourier space involves $(k-1)$-fold convolutions of $\widehat{u}$, where $\widehat{\bar{u}}(\xi) = \overline{\widehat{u}(-\xi)}$. Since S^{n-1} is symmetric under $\xi \mapsto -\xi$, the support of all convolved terms lies in the sumset of copies of $\pm S^{n-1}$, and multiplication by σ restricts back to S^{n-1}. □

Remark 4.2. In dimension $n \geq 2$, the conclusion $\operatorname{supp}\widehat{u} \subseteq S^{n-1}$ means u is a *Herglotz wave function*, satisfying $(-\Delta - 1)u = 0$.

4.1 Reduction to a density equation on the sphere

Write $\widehat{u} = \rho\, d\sigma$ for $\rho \in L^2(S^{n-1})$, so

$$u(x) = \int_{S^{n-1}} \rho(\omega) e^{ix\cdot\omega}\, d\sigma(\omega).$$

The equation $u = c(u^k * \widehat{\sigma})$ then reduces to the *spherical fixed-point equation*

$$\rho(\omega) = c \int_{V_k(\omega)} \rho(\omega_1) \cdots \rho(\omega_k)\, d\sigma_{\text{fiber}}(\omega_1, \ldots, \omega_k), \qquad \sigma\text{-a.e. } \omega \in S^{n-1}, \tag{3}$$

where $V_k(\omega) = \{(\omega_1, \ldots, \omega_k) \in (S^{n-1})^k : \omega_1 + \cdots + \omega_k = \omega\}$ is the incidence fibre.

Warning 4.3 (Incidence fibre $\neq$ spherical convolution). The operator $T(\rho)(\omega) = \int_{V_k(\omega)} \rho^{\otimes k}\, d\sigma_{\text{fiber}}$ is *not* the same as convolution by the spherical measure. In particular, the Funk–Hecke theorem does not diagonalise T in spherical harmonics. Any reduction to finitely many harmonic coefficients requires separate justification.

5 The One-Dimensional Case: Complete Solution

For $n = 1$, $S^0 = \{-1, +1\}$, $\sigma = \delta_{-1} + \delta_{+1}$, and $f(x) = \widehat{\sigma}(x) = 2\cos x$. Proposition 4.1 forces $\widehat{u} = a\delta_{-1} + b\delta_{+1}$, so $u(x) = ae^{-ix} + be^{ix}$.

5.1 Algebraic reduction

Expanding u^k and computing $u^k * f$ picks out frequencies ± 1, which exist only when k is **odd**.

Proposition 5.1 (Parity obstruction)**.** *If k is even, the equation $u = c(u^k * f)$ with $f = 2\cos$ has only $u \equiv 0$.*

For k odd, setting $\beta_k := \binom{k}{(k-1)/2}$ and comparing coefficients of $e^{\mp ix}$ gives the single constraint

$$ab = \lambda_k, \qquad \lambda_k := \left(\frac{1}{2\pi c \beta_k}\right)^{2/(k-1)}. \tag{4}$$

Theorem 5.2 (Complete solution in 1D)**.** ***[Proved]*** *Let $k \geq 3$ be odd, $c \in \mathbb{C} \setminus \{0\}$, $f = 2\cos$. The solution set of $u = c(u^k * f)$ in the Paley–Wiener class is*

$$\{0\} \cup \bigsqcup_{j=0}^{(k-3)/2} \{(a,b) \in \mathbb{C}^2 : ab = \omega^j \lambda_k\}, \qquad \omega = e^{4\pi i/(k-1)}.$$

This is a finite union of $(k-1)/2$ smooth one-dimensional complex manifolds (each $\cong \mathbb{C}^*$), plus the origin. Each component is exactly the translation orbit of any single non-trivial solution on it: the translation $u(x) \mapsto u(x-t)$ acts by $(a,b) \mapsto (e^{-it}a, e^{it}b)$, preserving ab.

Example 5.3 ($k = 3$)**.** $\beta_3 = 3$. Constraint: $ab = (6\pi c)^{-1}$. One component, the translation orbit of $u(x) = (6\pi c)^{-1/2} \cos x$.

Example 5.4 ($k = 5$)**.** $\beta_5 = 10$. Constraint: $ab = (20\pi c)^{-1/2}$. Two components (the scaling group $\mu_2 = \{1, -1\}$ generates a second hyperbola).

Remark 5.5 (Real solutions)*.* For real-valued solutions $b = \bar{a}$, so $ab = |a|^2 \geq 0$. Only the hyperbola with $\lambda_k \in \mathbb{R}_{>0}$ contributes; the real solution manifold is the circle $|a| = |\lambda_k|^{1/2}$, parametrised by translations.

Part II
The S^1 Restriction Problem

6 The Sharp Restriction Constant and Its Euler–Lagrange Equation

6.1 Setup

The sharp Fourier extension constant on $S^1 \subset \mathbb{R}^2$ is

$$\mathcal{R} := \sup_{\substack{f \in L^2(S^1) \\ f \neq 0}} \frac{\|Ef\|_{L^6(\mathbb{R}^2)}}{\|f\|_{L^2(S^1)}}, \qquad (Ef)(x) := \int_{S^1} f(\omega) e^{ix\cdot\omega}\, d\sigma(\omega),$$

where σ is the arclength measure on S^1 (total mass 2π). Every extension $u = Ef$ satisfies $(-\Delta - 1)u = 0$.

6.2 The Euler–Lagrange equation

Normalise $\|f\|_{L^2(S^1)} = 1$. A critical point of $f \mapsto \|Ef\|_{L^6}^6$ on the unit sphere satisfies

$$E^*(|u|^4 u) = \Lambda f, \qquad u = Ef, \qquad \Lambda = \|u\|_{L^6}^6\,, \tag{5}$$

where $E^* : L^{6/5}(\mathbb{R}^2) \to L^2(S^1)$ is the adjoint restriction: $(E^*g)(\omega) = \int_{\mathbb{R}^2} g(x)e^{-ix\cdot\omega}\,dx$ restricted to $\omega \in S^1$. Equivalently,

$$u = c\big((|u|^4 u) * \widehat{\sigma}\big), \qquad c = \Lambda^{-1}. \tag{6}$$

Warning 6.1 (The nonlinearity is $|u|^4u$, not u^5). The restriction problem gives $|u|^4u = u^3\bar{u}^2$, which is degree 5 but *not* holomorphic. The linearisation is *real-linear*, not complex-linear: the Jacobian couples the Fourier mode n to mode $-n$. This distinction is essential for the spectral analysis in Section 8.

6.3 The symmetry group

The natural symmetry group is

$$G = U(1) \times SO(2) \ltimes \mathbb{R}^2, \qquad \dim G = 4.$$

It acts on densities by

$$(e^{i\alpha}, R, x_0) \cdot f(\omega) = e^{i\alpha}e^{-ix_0\cdot\omega}f(R^{-1}\omega).$$

Phase, rotation, and modulation/translation all preserve $\|f\|_{L^2}$ and $\|Ef\|_{L^6}$, hence preserve Λ. Every critical point generates a G-orbit of critical points with the same Λ.

Definition 6.2 (Shape)**.** A *shape* is an equivalence class of normalised critical points modulo the G-action. Two critical points with the same shape have the same critical value Λ.

6.4 What "finding all critical values" means

The search problem is: determine the finite set

$$\mathcal{Z} := \{(f, \Lambda) : E^*(|Ef|^4 Ef) = \Lambda f,\ \|f\|_2 = 1\}/G.$$

If $\mathcal{Z}$ is finite, the sharp restriction constant is

$$\mathcal{R} = \max_{[f,\Lambda]\in\mathcal{Z}} \Lambda^{1/6} = c_{\min}^{-1/6},$$

where $c_{\min}$ is the smallest positive $c_j = \Lambda_j^{-1}$. The normalisation $\|f\|_2 = 1$ is essential: without it, αu solves the equation with c replaced by $c\alpha^{-4}$, preventing discreteness.

6.5 Positive representatives for extremisers

The exponent 6 gives a useful reduction for the *extremiser* search.

Lemma 6.3 (Even-exponent positivity reduction)**.** *For every $f \in L^2(S^1)$,*

$$\|Ef\|_{L^6(\mathbb{R}^2)} \leq \|E|f|\|_{L^6(\mathbb{R}^2)}.$$

Consequently, if f is an extremiser, then $|f|$ is also an extremiser. In particular, to identify the sharp constant it is enough to search among nonlinear eigenfunctions with a nonnegative representative.

Proof. By Plancherel and the fact that $6 = 2 \cdot 3$, up to the fixed Fourier normalisation constant,

$$\|Ef\|_{L^6}^6 = \|(f\sigma) * (f\sigma) * (f\sigma)\|_{L^2(\mathbb{R}^2)}^2.$$

The convolution density satisfies the pointwise estimate

$$|(f\sigma) * (f\sigma) * (f\sigma)| \leq (|f|\sigma) * (|f|\sigma) * (|f|\sigma)$$

almost everywhere. Squaring and integrating, then applying Plancherel in the reverse direction, gives the asserted inequality.

Since $\||f|\|_{L^2(S^1)} = \|f\|_{L^2(S^1)}$, replacing an extremiser by its modulus preserves the quotient and hence produces another extremiser. □

Remark 6.4. Lemma 6.3 is an extremiser reduction, not a classification of all critical points. A non-maximal critical point need not have a nonnegative representative. Thus the full compactness theorem for every critical sequence remains stronger than what is needed merely to compute $\mathcal{R}$.

7 Fredholm Structure of the Linearisation

7.1 The extension operator E is not compact

One might hope that $E : L^2(S^1) \to L^6(\mathbb{R}^2)$ is compact, which would immediately give compactness of the nonlinear map $T(f) = E^*(|Ef|^4 Ef)$. This is **false**.

Proposition 7.1 (E is not compact)**.** ***[Proved]*** *The extension operator $E : L^2(S^1) \to L^6(\mathbb{R}^2)$ is bounded (Stein–Tomas) but not compact.*

Proof. Fix $f \in L^2(S^1)$ with $\|f\|_2 = 1$ and $\|Ef\|_{L^6} > 0$. For $n \in \mathbb{N}$, define $f_n(\omega) := e^{-ine_1 \cdot \omega} f(\omega)$ where $e_1 = (1, 0)$. Then $\|f_n\|_2 = 1$ and $Ef_n(x) = Ef(x - ne_1) = u(x - ne_1)$, a translate of $u = Ef$.

For $n \neq m$ with $|n - m|$ large, the supports of $|u(\cdot - ne_1)|$ and $|u(\cdot - me_1)|$ in any L^6-significant region are essentially disjoint (since $u(x) \to 0$ as $|x| \to \infty$ by Riemann–Lebesgue), so

$$\|Ef_n - Ef_m\|_{L^6} = \|u(\cdot - ne_1) - u(\cdot - me_1)\|_{L^6} \xrightarrow{|n-m|\to\infty} 2^{1/6} \|u\|_{L^6} > 0.$$

Hence $\{Ef_n\}$ has no Cauchy subsequence. □

The obstruction is the noncompact translation symmetry: high-frequency modulations on S^1 shift the extension arbitrarily far in x-space. This means compactness of the critical set requires quotienting by G; it cannot be deduced from a single operator bound.

7.2 The linearised operator K_{u_0} is compact

Despite the non-compactness of E, the *specific* operator appearing in the linearisation is compact, thanks to the decay of the solution u_0.

Definition 7.2. At a critical point f_0 with $u_0 = Ef_0$, define the linearised operator $K = K_{u_0} : L^2(S^1) \to L^2(S^1)$ by

$$Kh := E^*\big(DN(u_0)[Eh]\big),$$

where $DN(u_0)[v] = 3|u_0|^4 v + 2|u_0|^2 u_0^2 \bar{v}$ is the (real-linear) Fréchet derivative of $N(u) = |u|^4 u$.

Theorem 7.3 (Compactness of K_{u_0})**.** ***[Proved]*** *For any $u_0 = Ef_0$ with $f_0 \in L^2(S^1)$, the operator $K_{u_0} : L^2(S^1) \to L^2(S^1)$ is compact.*

Proof. We prove compactness for the "main term" $K_1 h := E^*(|u_0|^4 Eh)$; the argument for $K_2 h := E^*(|u_0|^2 u_0^2 \overline{Eh})$ is identical, and $K = 3K_1 + 2K_2$.

Step 1: Truncation. For $R > 0$ let χ_R denote the indicator of $B(0,R) \subset \mathbb{R}^2$ and set $K_1^R h := E^*(\chi_R\, |u_0|^4\, Eh)$. Then

$$\big\|K_1 - K_1^R\big\|_{L^2 \to L^2} \le \|E^*\|_{L^{6/5} \to L^2}\ \big\|(1-\chi_R)|u_0|^4\big\|_{L^{3/2}}\ \|E\|_{L^2 \to L^6}\,.$$

Here $\|E\|$ and $\|E^*\|$ are the (finite) Stein–Tomas constants, and

$$\big\|(1-\chi_R)|u_0|^4\big\|_{L^{3/2}}^{3/2} = \int_{|x|>R} |u_0|^6\, dx \ \xrightarrow{R\to\infty}\ 0,$$

since $u_0 \in L^6(\mathbb{R}^2)$. Hence $\big\|K_1 - K_1^R\big\|_{\mathrm{op}} \to 0$.

Step 2: K_1^R is Hilbert–Schmidt. The integral kernel of K_1^R on $S^1 \times S^1$ is

$$\mathcal{K}_R(\omega, \omega') = \int_{B(0,R)} |u_0(x)|^4\, e^{ix\cdot(\omega'-\omega)}\, dx.$$

Since $|u_0|^4 \chi_R \in L^1(\mathbb{R}^2) \cap L^\infty(\mathbb{R}^2)$ (bounded and compactly supported), its Fourier transform is continuous on $\mathbb{R}^2$. Therefore $\mathcal{K}_R$ is continuous on the compact set $S^1 \times S^1$, hence bounded, and in particular belongs to $L^2(S^1 \times S^1)$. Thus K_1^R is Hilbert–Schmidt, hence compact.

Step 3: Conclusion. $K_1 = \lim_{R\to\infty} K_1^R$ in operator norm, where each K_1^R is compact. Hence K_1 is compact. □

Corollary 7.4 (Fredholm structure)**.** ***[Proved]*** *At any critical point f_0 of the restriction functional with $\|f_0\|_2 = 1$ and $\Lambda_0 = \|Ef_0\|_6^6 > 0$, the operator*

$$L := K_{u_0} - \Lambda_0 I$$

has the form $L = K - \Lambda_0 I$ with K compact. Therefore $L : L^2(S^1) \to L^2(S^1)$ is Fredholm of index zero. In particular, $\ker L$ *is finite-dimensional, and L is invertible if and only if* $\ker L = \{0\}$.

Remark 7.5 (Self-adjointness). With respect to the real inner product $\operatorname{Re}\langle f, g\rangle_{L^2(S^1)}$, the operator K is self-adjoint. Indeed, $DN(u_0)$ is self-adjoint on $L^2(\mathbb{R}^2, \operatorname{Re}\langle\cdot,\cdot\rangle)$ (it is the Hessian of the real functional $u \mapsto \frac{1}{6}|u|^6$), and E^* is the adjoint of E. Consequently, the eigenvalues of L are real.

8 Nondegeneracy at the Constant Solution

8.1 Setup

The constant function $f_0 = (2\pi)^{-1/2}$ has $\|f_0\|_{L^2(S^1)} = 1$. Its extension is the radial function

$$u_0(x) = Ef_0(x) = (2\pi)^{1/2} J_0(|x|),$$

where J_0 is the Bessel function of order zero. Set $A := (2\pi)^{1/2}$, so $u_0(r) = AJ_0(r)$. The critical value is

$$\Lambda = \|u_0\|_{L^6}^6 = A^6 \cdot 2\pi \int_0^\infty J_0(r)^6\, r\, dr = 16\pi^4 \int_0^\infty J_0(r)^6\, r\, dr.$$

The stabiliser of f_0 under G is $\operatorname{Stab}_G(f_0) = SO(2)$ (rotations of S^1 fix the constant function). The orbit has dimension $\dim G - \dim \operatorname{Stab} = 4 - 1 = 3$, generated by phase ($U(1)$) and translations ($\mathbb{R}^2$).

8.2 Fourier decomposition of the linearised operator

Since u_0 is radial, the linearised operator $K : L^2(S^1) \to L^2(S^1)$ preserves each pair of Fourier modes $\{e^{in\theta}, e^{-in\theta}\}$. We use the Fourier expansion

$$h(\theta) = \sum_{m\in\mathbb{Z}} c_m e^{im\theta}, \qquad Eh(r,\phi) = \sum_{m\in\mathbb{Z}} 2\pi\, i^{|m|} c_m\, J_{|m|}(r)\, e^{im\phi},$$

where (r, ϕ) are polar coordinates in $\mathbb{R}^2$.

Proposition 8.1 (Eigenvalue structure)**.** ***[Proved]*** *Define the Bessel integrals*

$$\mathcal{J}_n := \int_0^\infty J_0(r)^4\, J_n(r)^2\, r\, dr, \qquad n \geq 0, \tag{7}$$

and $\mathcal{I}_n := 16\pi^4 \mathcal{J}_n$*, so that* $\Lambda = \mathcal{I}_0$*. Then, on the sector* $\{c_n e^{in\theta} + c_{-n} e^{-in\theta}\}$ *with* $n \geq 1$*, the operator* $K = E^* DN(u_0) E$ *acts as a* 4×4 *real-linear map with eigenvalues*

$$5\mathcal{I}_n \quad (\textit{multiplicity } 2) \qquad \textit{and} \qquad \mathcal{I}_n \quad (\textit{multiplicity } 2).$$

On the sector $n = 0$*, the eigenvalues are* $5\mathcal{I}_0$ *(for real perturbations) and* $\mathcal{I}_0$ *(for imaginary perturbations), each with multiplicity* 1*.*

Proof. Write $v = Eh$ where $h = c_n e^{in\theta} + c_{-n} e^{-in\theta}$. Then $v = 2\pi i^n J_n(r)[c_n e^{in\phi} + c_{-n} e^{-in\phi}]$ (using $E(e^{im\theta}) = 2\pi i^{|m|} J_{|m|}(r) e^{im\phi}$).

Since $u_0 = AJ_0(r)$ is real and radial, $DN(u_0)[v] = 3u_0^4 v + 2u_0^4 \bar{v}$. The $e^{in\phi}$ component of $DN(u_0)[v]$ is $2\pi A^4 J_0^4 J_n[3i^n c_n + 2(-i)^n \bar{c}_{-n}]$, and the $e^{-in\phi}$ component is $2\pi A^4 J_0^4 J_n[3i^n c_{-n} + 2(-i)^n \bar{c}_n]$.

Applying E^* and using the identity $\langle E^* g, e^{im\theta} \rangle = 2\pi(-i)^{|m|} \int_0^\infty g_m(r) J_{|m|}(r)\, r\, dr$ (where g_m is the m-th angular Fourier coefficient of g), together with $(-i)^n i^n = 1$ and $(-i)^{2n} = (-1)^n$, one computes:

$$(Kh)_n = \mathcal{I}_n \big[3c_n + 2(-1)^n \bar{c}_{-n}\big], \tag{8}$$

$$(Kh)_{-n} = \mathcal{I}_n \big[3c_{-n} + 2(-1)^n \bar{c}_n\big]. \tag{9}$$

Writing $c_n = p + iq$, $c_{-n} = s + it$ (real and imaginary parts), the system decouples into a *real part* and an *imaginary part*, each a 2×2 real system. For both parities of n, both 2×2 matrices have the same eigenvalues: 5 and 1 (with the eigenvectors exchanging between the real and imaginary blocks as $(-1)^n$ changes sign). Hence the eigenvalues of K on the sector are $5\mathcal{I}_n$ and $\mathcal{I}_n$, each with multiplicity 2. The $n = 0$ case is analogous with multiplicities halved. □

8.3 Identification of the symmetry kernel

Proposition 8.2 (Symmetry directions)**.** ***[Proved]*** *The kernel of* $L = K - \Lambda I$ *on* $L^2(S^1)$ *includes the following directions, with the indicated eigenvalue identities.*

1. ***Phase*** *(*$n = 0$*, imaginary):* $h = if_0$*. This lies in the* $\mathcal{I}_0$*-eigenspace of* K*. Since* $\Lambda = \mathcal{I}_0$*, the* L*-eigenvalue is* 0*.*

2. ***Translations*** *(*$n = 1$*):* $h = -i(\hat{x}_0 \cdot \omega) f_0$ *for* $\hat{x}_0 \in \mathbb{R}^2$*. This lies in the* $5\mathcal{I}_1$*-eigenspace of* K*. Since translations are an exact symmetry,* $5\mathcal{I}_1 = \Lambda$*.*

3. ***No rotation direction****:* $\partial_\theta f_0 = 0$ *since* f_0 *is constant.*

In particular, $\mathcal{I}_1 = \Lambda/5$*, i.e.,*

$$5 \int_0^\infty J_0(r)^4\, J_1(r)^2\, r\, dr = \int_0^\infty J_0(r)^6\, r\, dr. \tag{10}$$

This is a Bessel function identity forced by translation invariance.

Proof. (i) is verified by direct computation: $E(if_0) = iu_0$, $DN(u_0)[iu_0] = 3u_0^4(iu_0) + 2u_0^4(-iu_0) = iu_0^5$, and $E^*(iu_0^5) = iE^*(u_0^5) = i\Lambda f_0 = \Lambda(if_0)$.

(ii) The translation $f \mapsto e^{-ix_0\cdot\omega}f$ gives $h = -i(\hat{x}_0 \cdot \omega)f_0$ at first order. The corresponding extension perturbation is $Eh = -\hat{x}_0 \cdot \nabla u_0$ (verified by differentiating $E(e^{-ix_0\cdot\omega}f_0)(x) = u_0(x - x_0)$ in x_0). Differentiating the Euler–Lagrange equation $E^*(|u_0(\cdot - x_0)|^4 u_0(\cdot - x_0)) = \Lambda e^{-ix_0\cdot\omega}f_0$ in x_0 at $x_0 = 0$ gives $Kh = \Lambda h$, confirming that h is in $\ker L$.

From the eigenvalue structure, $h = -i(\hat{x}_0 \cdot \omega)f_0$ has Fourier support on $n = \pm 1$ with $c_{-1} = -\bar{c}_1$. Substituting into (8)–(9) shows this lies in the $5\mathcal{I}_1$-eigenspace, whence $5\mathcal{I}_1 = \Lambda$.

(iii) f_0 is constant, so $\partial_\theta f_0 = 0$. □ □

Remark 8.3 (Euler identity). The identity $Kf_0 = 5\Lambda f_0$ (in the $n = 0$ real sector) follows from the degree-5 homogeneity of N: $DN(u)[u] = 5N(u)$ for all u. Hence $Kf_0 = E^*(DN(u_0)[u_0]) = 5E^*(N(u_0)) = 5\Lambda f_0$.

8.4 The nondegeneracy criterion

The eigenvalues of $L = K - \Lambda I$ on the n-th sector are $5\mathcal{I}_n - \Lambda$ and $\mathcal{I}_n - \Lambda$. At $n = 0$: 0 (phase) and 4Λ (amplitude). At $n = 1$: 0 (translations, multiplicity 2) and $-4\Lambda/5$ (multiplicity 2).

Theorem 8.4 (Nondegeneracy reduces to Bessel integrals)**.** ***[Conditional]*** *The constant solution $f_0 = (2\pi)^{-1/2}$ is nondegenerate modulo G (i.e., $\ker L = T_{f_0}(G \cdot f_0)$ on the tangent space to $\|f\|_2 = 1$) if and only if*

$$\boxed{\mathcal{I}_n \neq \Lambda \quad \text{and} \quad 5\mathcal{I}_n \neq \Lambda \qquad \text{for all } n \geq 2.} \tag{11}$$

Equivalently, in terms of the Bessel integrals (7)*:*

$$\mathcal{J}_n \neq \mathcal{J}_0 \quad \text{and} \quad 5\mathcal{J}_n \neq \mathcal{J}_0 \qquad \text{for all } n \geq 2.$$

Proof. By Proposition 8.1, the eigenvalues of L on the n-th sector ($n \geq 2$) are $5\mathcal{I}_n - \Lambda$ and $\mathcal{I}_n - \Lambda$, each with multiplicity 2. These are both nonzero if and only if (11) holds. The symmetry directions exhaust the kernel at $n = 0$ and $n = 1$ by Proposition 8.2. □

8.5 Asymptotic analysis of $\mathcal{J}_n$

Lemma 8.5 (Decay of $\mathcal{J}_n$)**.** ***[Proved]*** *$\mathcal{J}_n \to 0$ as $n \to \infty$. More precisely, $\mathcal{J}_n = O(n^{-1})$.*

Proof. For $r < n(1 - \varepsilon)$, the Bessel function $J_n(r)$ is exponentially small. For $r \geq n$, the uniform bound $|J_0(r)| \leq 1$ and the asymptotic $|J_\nu(r)| \leq Cr^{-1/2}$ (valid for $r \geq \nu$ and all $\nu \geq 0$) give

$$\int_n^\infty J_0(r)^4\, J_n(r)^2\, r\, dr \leq C \int_n^\infty r^{-2} \cdot r^{-1} \cdot r\, dr = C \int_n^\infty r^{-2}\, dr = C/n. \qquad \square$$

Since $\mathcal{J}_n \to 0$ while $\mathcal{J}_0 > 0$ and $\mathcal{J}_0/5 > 0$, the conditions (11) hold for all sufficiently large n. Only finitely many values of n need to be checked.

Lemma 8.6 (Parity structure of the asymptotics)**.** *For moderate n (before the J_n-support retreats past the main lobe of J_0), the large-r average of $J_0(r)^4 J_n(r)^2$ depends on the parity of n. Using the asymptotics $J_\nu(r) \sim \sqrt{2/\pi r}\cos(r - \nu\pi/2 - \pi/4)$ for $r \gg \nu$:*

- ***Even** n:* $\cos(r - n\pi/2 - \pi/4) = \pm\cos(r - \pi/4)$*, so* $\langle J_0^4 J_n^2\rangle_{\text{avg}} \sim (2/\pi r)^3 \cdot \frac{5}{16}$*, the same coefficient as* $\langle J_0^6\rangle$*.*
- ***Odd** n:* $\cos(r - n\pi/2 - \pi/4) = \pm\sin(r - \pi/4)$*, so* $\langle J_0^4 J_n^2\rangle_{\text{avg}} \sim (2/\pi r)^3 \cdot \frac{1}{16}$*, which is* $\frac{1}{5}$ *of the even case.*

Consequently, for moderate even n, $\mathcal{J}_n$ is close to $\mathcal{J}_0$ (deficiency only from the small-r region where $J_n(r) \ll J_0(r)$), while for moderate odd n, $\mathcal{J}_n$ is close to $\mathcal{J}_0/5 = \mathcal{J}_1$.

Remark 8.7 (Which conditions are tight?). The identity $\mathcal{J}_1 = \mathcal{J}_0/5$ (from translation invariance) shows that the condition $5\mathcal{J}_n = \mathcal{J}_0$ *is* achieved at $n = 1$. For even n, the condition $\mathcal{J}_n = \mathcal{J}_0$ is never achieved ($\mathcal{J}_n < \mathcal{J}_0$ due to the small-r deficiency), but the margin may be small for $n = 2$. The most delicate check is therefore whether $5\mathcal{J}_n = \mathcal{J}_0$ for some *even* $n \geq 2$ (where $\mathcal{J}_n$ is close to $\mathcal{J}_0$, hence $5\mathcal{J}_n$ is close to $5\mathcal{J}_0 \gg \Lambda$, making this condition safe), and whether $\mathcal{J}_n = \mathcal{J}_0$ for some *odd* $n \geq 3$ (where $\mathcal{J}_n \approx \mathcal{J}_0/5$, well below $\mathcal{J}_0$, also safe).

Upshot: the asymptotic structure makes the nondegeneracy conditions (11) very plausible, but does not close the argument. A rigorous verification requires evaluating $\mathcal{J}_n$ for $n = 2, 3, \ldots, n_*$ (where n_* is the threshold beyond which the $O(1/n)$ bound suffices) and confirming the two inequalities at each. This is a finite numerical computation that can be made rigorous with interval arithmetic.

8.6 Nondegeneracy at non-constant solutions

If there exist non-constant critical points of the restriction functional, their nondegeneracy must be established separately. At a non-constant f_0, the extension $u_0 = Ef_0$ is not radial, so K_{u_0} does not decompose into Fourier sectors. Two approaches are available:

Direct spectral analysis. If u_0 has a discrete symmetry (e.g., if f_0 is a real-valued even function on S^1), the problem decomposes into finitely many symmetry sectors, each finite-dimensional.

Generic transversality (Sard–Smale). Introduce a family of G-equivariant perturbations $F_p(\rho) = \rho - cT_p(\rho)$ and show that the parameter derivatives span the cokernel of $DF_p(\rho)$ at every zero. This gives nondegeneracy for a generic parameter p. To cover the original (unperturbed) problem, one must either verify that the spherical measure is a regular parameter, or continue the generic solution set to the original parameter without bifurcation.

Remark 8.8 (Varying c alone is insufficient). As noted in Warning 4.3, varying only the scalar c provides a one-parameter family, which is generically too small for Sard–Smale (the cokernel at a degenerate zero may have dimension > 1). A richer family of perturbations is needed.

9 Compactness Modulo Symmetry

9.1 Regularity bootstrap

Proposition 9.1 (Smoothness and gauge-fixed uniform bounds)**.** ***[Proved]*** *If $f \in L^2(S^1)$ satisfies $E^*(|Ef|^4 Ef) = \Lambda f$ with $\Lambda > 0$, then $f \in C^\infty(S^1)$. Moreover, for each $k \geq 0$ there exists $C_k = C_k(\varepsilon)$ such that $\|f\|_{C^k(S^1)} \leq C_k$ for all normalised critical points with $\Lambda \geq \varepsilon$ in a fixed translation gauge; in particular, this applies in the nonnegative slice used in Lemma 9.11.*

Proof sketch. Write $u = Ef$, $g = |u|^4 u$. Since $\widehat{u} = f\, d\sigma$ is supported on S^1, the Fourier transform $\widehat{g}$ is supported in the Minkowski sum $S^1 + S^1 + S^1 + S^1 + S^1 = \overline{B(0,5)}$ (using $|u|^4 u = u^3 \bar{u}^2$ and $\widehat{\bar{u}}$ supported on $-S^1 = S^1$). Thus g is *frequency-localised* to $B(0,5)$.

The Euler–Lagrange equation gives $f = \Lambda^{-1} \widehat{g}\big|_{S^1}$, so f inherits the regularity of $\widehat{g}$ restricted to the compact curve $S^1 \subset B(0,5)$. The n-th Fourier coefficient of f on S^1 is

$$\widehat{f}(n) = \Lambda^{-1} \cdot 2\pi(-i)^{|n|} \int_0^\infty g_n(r)\, J_{|n|}(r)\, r\, dr, \tag{12}$$

where $g_n(r)$ is the n-th angular Fourier coefficient of g. For $|n| \gg 1$, $J_{|n|}(r)$ is exponentially small for $r < |n|/2$ (Debye bounds), and the contribution from $r > |n|/2$ is controlled by the $L^{6/5}$ tail of g via Hölder and the Bessel asymptotics $J_\nu(r) = O(r^{-1/2})$ for $r > \nu$. An inductive argument (improved decay of $\widehat{f}(n)$ feeds back into improved regularity of f and hence of u and g) yields $|\widehat{f}(n)| = O(|n|^{-N})$ for all N. The constants are obtained from the L^2 normalisation, the lower bound $\Lambda \geq \varepsilon$, and the Euler–Lagrange equation, by this energy/bootstrap scheme.

The translation-gauge qualification is necessary. If $f_x(\omega) = e^{-ix\cdot\omega} f(\omega)$, then f_x is again a normalised critical point with the same Λ, while $\|f_x\|_{C^1}$ grows like $|x|$ in general. Once the translation parameter is fixed, or once one works in the nonnegative slice below, this obstruction is absent. □

Corollary 9.2 (Uniform pointwise decay in a fixed gauge)**.** *For all normalised critical points in a fixed translation gauge with $\Lambda \geq \varepsilon$,*

$$|Ef(x)| \leq C(\varepsilon)\,(1+|x|)^{-1/2} \qquad \text{for all } x \in \mathbb{R}^2.$$

Equivalently, for an arbitrary translated representative the same estimate holds around its translation centre.

Proof. By the bootstrap, $\|f\|_{C^1} \leq C(\varepsilon)$. The standard stationary-phase estimate for extension from a C^∞ curve in $\mathbb{R}^2$ with C^1 density gives $|Ef(x)| = O(|x|^{-1/2})$ for $|x| \geq 1$, and $|Ef(x)| \leq (2\pi)^{1/2}\,\|f\|_{L^2} \leq (2\pi)^{1/2}$ everywhere. □

9.2 Profile decomposition for the circle extension

The extension $E : L^2(S^1) \to L^6(\mathbb{R}^2)$ is not compact (Proposition 7.1), and the sole obstruction is the translation symmetry. The following decomposition, analogous to results of Merle–Vega and Shao for the paraboloid, captures this defect of compactness completely.

Theorem 9.3 (Profile decomposition)**.** ***[Conditional]*** *Let $f_j \in L^2(S^1)$ with $\|f_j\|_2 \leq C$. After passing to a subsequence, there exist profiles $\phi^k \in L^2(S^1)$ $(k = 1, 2, \ldots)$ and group elements $g_j^k = (e^{i\alpha_j^k}, R_j^k, x_j^k) \in G$ satisfying the* divergence condition *$|x_j^k - x_j^l| \to \infty$ for all $k \neq l$, such that for each $K \geq 1$,*

$$f_j = \sum_{k=1}^{K} (g_j^k)^{-1} \cdot \phi^k + r_j^K, \tag{13}$$

with the decoupling identities

$$\|f_j\|_2^2 = \textstyle\sum_{k=1}^{K} \left\|\phi^k\right\|_2^2 + \left\|r_j^K\right\|_2^2 + o(1), \tag{14}$$

$$\|Ef_j\|_6^6 = \textstyle\sum_{k=1}^{K} \left\|E\phi^k\right\|_6^6 + \left\|Er_j^K\right\|_6^6 + o(1), \tag{15}$$

and $\left\|Er_j^K\right\|_{L^6} \to 0$ as $K \to \infty$ (after further subsequence extraction at each stage).

The L^2 decoupling (14) is elementary: for $k \neq l$, $\left\langle e^{-ix_j^k\cdot\omega}\phi^k,\, e^{-ix_j^l\cdot\omega}\phi^l \right\rangle = \int_{S^1} \phi^k\overline{\phi^l}\, e^{i(x_j^l - x_j^k)\cdot\omega}\, d\sigma \to 0$ by the Riemann–Lebesgue lemma on S^1.

The L^6 decoupling (15) is deeper. Since $E(e^{-ix_j^k\cdot\omega}\phi^k)(x) = (E\phi^k)(x - x_j^k)$, distinct profiles produce extensions centred at diverging locations. A Brézis–Lieb argument using the Herglotz wave decay $|E\phi^k(x)| = O(|x|^{-1/2})$ (valid once ϕ^k is smooth; for L^2 profiles, one approximates and uses density) then gives the L^6 asymptotic additivity.

The *construction* of the decomposition proceeds iteratively. The key input is an inverse Strichartz inequality for S^1:

Lemma 9.4 (Inverse Strichartz for S^1)**.** *If $\|f_j\|_2 = 1$ and $\|Ef_j\|_{L^6} \geq \delta > 0$, then there exist $x_j \in \mathbb{R}^2$ such that the re-centred sequence $\tilde{f}_j(\omega) := e^{ix_j\cdot\omega} f_j(\omega)$ has a subsequence converging weakly to some $\phi \neq 0$ in $L^2(S^1)$.*

Proof. Since $\|Ef_j\|_6^6 \geq \delta^6$, a covering argument gives $x_j \in \mathbb{R}^2$ with $\int_{B(x_j,1)} |Ef_j|^6 \geq c\delta^6$. Set $\tilde{u}_j(x) := Ef_j(x + x_j) = E\tilde{f}_j(x)$. Then $\int_{B(0,1)} |\tilde{u}_j|^6 \geq c\delta^6$.

Every Herglotz wave satisfies $(\Delta + 1)u = 0$ and $\|u\|_{L^\infty} \leq (2\pi)^{1/2} \|f\|_2 = (2\pi)^{1/2}$. By interior Schauder estimates for $\Delta u + u = 0$, the family $\{\tilde{u}_j\}$ is equicontinuous on $\overline{B(0,1)}$. Arzelà–Ascoli extracts a uniformly convergent subsequence $\tilde{u}_j \to u_\infty$ on $\overline{B(0,1)}$.

Meanwhile, $\tilde{f}_j \rightharpoonup \phi$ (Banach–Alaoglu, after subsequence). Weak convergence gives $E\tilde{f}_j(x) \to E\phi(x)$ pointwise for every x (since $\omega \mapsto e^{ix\cdot\omega} \in L^2(S^1)$). So $u_\infty = E\phi$. From $\int_{B(0,1)} |u_\infty|^6 \geq c\delta^6 > 0$, we get $\phi \neq 0$. □

Remark 9.5 (Simplification from compact frequency support)*.* For the paraboloid $\{(\xi, |\xi|^2)\}$, profile decompositions must account for parabolic rescalings (a second family of symmetries). For S^1, the frequency set is compact with no scaling symmetry, so translations are the sole defect of compactness. Moreover, the inverse Strichartz lemma follows from elliptic regularity for $\Delta u + u = 0$, rather than from bilinear restriction estimates.

9.3 Profiles of critical sequences inherit the equation

The following result is the key structural input: if the original sequence consists of critical points, then each profile in the decomposition is itself a critical point.

Proposition 9.6 (Each profile is a critical point)**.** ***[Proved]*** *Let (f_j, Λ_j) be normalised critical points with $\Lambda_j \to \Lambda_\infty > 0$, and let $\phi^1, \phi^2, \ldots$ be the profiles from Theorem 9.3 (re-indexed so that $x_j^1 = 0$). Then each profile satisfies*

$$E^*(|E\phi^k|^4 E\phi^k) = \Lambda_\infty \phi^k. \tag{16}$$

In particular, $\psi^k := \phi^k / \left\|\phi^k\right\|_2$ is a normalised critical point with eigenvalue $\widetilde{\Lambda}_k = \Lambda_\infty / \left\|\phi^k\right\|_2^4$.

Proof. We prove (16) for $k = 1$; the case $k \geq 2$ follows by applying the translation $y_j^k := x_j^k - x_j^1$ (which diverges) and repeating the argument.

Set $v^l := E\phi^l$ and $w_j := Er_j$. Write $\tilde{f}_j := e^{ix_j^1\cdot\omega} f_j$ for the re-centred sequence, so $\tilde{u}_j := E\tilde{f}_j = v^1 + \sum_{l\geq 2} v^l(\cdot - y_j^l) + w_j$. The Euler–Lagrange equation, tested against arbitrary $h \in L^2(S^1)$, reads

$$\Lambda_j \left\langle \tilde{f}_j,\, h \right\rangle_{L^2(S^1)} = \int_{\mathbb{R}^2} |\tilde{u}_j|^4 \tilde{u}_j \, \overline{Eh} \, dx. \tag{17}$$

Left side. Since $\tilde{f}_j = \phi^1 + \sum_{l\geq 2} e^{-iy_j^l\cdot\omega}\phi^l + r_j$ and $\left\langle e^{-iy_j^l\cdot\omega}\phi^l,\, h\right\rangle \to 0$ by Riemann–Lebesgue ($|y_j^l| \to \infty$), the left side converges to $\Lambda_\infty \langle \phi^1,\, h\rangle$.

Right side. Decompose $|\tilde{u}_j|^4\tilde{u}_j = |v^1|^4 v^1 + N_j$, where N_j collects all terms containing at least one factor of $\eta_j := \tilde{u}_j - v^1 = \sum_{l\geq 2} v^l(\cdot - y_j^l) + w_j$. We show $\int N_j\, \overline{Eh} \to 0$.

Step (i): η_j vanishes locally in L^6. For each fixed $R > 0$: $\left\|v^l(\cdot - y_j^l)\right\|_{L^6(B(0,R))} = \left\|v^l\right\|_{L^6(B(-y_j^l,R))} \to 0$ as $j \to \infty$ (since $v^l \in L^6(\mathbb{R}^2)$ and $B(-y_j^l, R)$ escapes to infinity), and $\|w_j\|_{L^6} \to 0$. Hence $\|\eta_j\|_{L^6(B(0,R))} \to 0$.

Step (ii): individual translated-profile terms vanish. Fix $l \geq 2$. The typical cross term to estimate is $I_j^l := \int |v^1(x)|^4\, v^l(x - y_j^l)\, \overline{Eh(x)}\, dx$. Split $\mathbb{R}^2$ into $B(0,R)$ and its complement.

- On $B(0,R)$: $|v^1| \leq (2\pi)^{1/2}$, so $|I_j^l|_{B(0,R)}| \leq C \left\|v^l(\cdot - y_j^l)\right\|_{L^6(B(0,R))} \|Eh\|_{L^6} \to 0$ by step (i).
- On $\mathbb{R}^2 \setminus B(0,R)$: $|I_j^l|_{\mathbb{R}^2\setminus B}| \leq C \left\|v^l\right\|_6 \|Eh\|_{L^6(|x|>R)} \to 0$ as $R \to \infty$ (since $Eh \in L^6$ has vanishing tails).

An $\varepsilon/3$ argument in R and j gives $I_j^l \to 0$.

Step (iii): the remainder. Terms involving $w_j = Er_j$ are bounded by $C\, \|w_j\|_6\, \|Eh\|_6 \to 0$ (using $\|v^1\|_\infty \leq C$ and Hölder).

Step (iv): higher-order cross terms. Every monomial in N_j has at least one factor η_j or $\bar{\eta}_j$. The same local-vanishing argument applies: on $B(0,R)$, the factor of η_j is small in L^6; on the complement, Eh is small in L^6.

Combining: the right side of (17) converges to $\int |v^1|^4 v^1\, \overline{Eh} = \langle E^*(|E\phi^1|^4 E\phi^1),\, h\rangle$. Since h is arbitrary, $E^*(|E\phi^1|^4 E\phi^1) = \Lambda_\infty \phi^1$.

The rescaling: if $\alpha_k := \left\|\phi^k\right\|_2 > 0$, then $\psi^k := \phi^k/\alpha_k$ satisfies $E^*(|E\psi^k|^4 E\psi^k) = \alpha_k^{-5} E^*(|E\phi^k|^4 E\phi^k) = \Lambda_\infty \alpha_k^{-5}\phi^k = (\Lambda_\infty/\alpha_k^4)\psi^k$. □

9.4 Existence of a maximiser

Theorem 9.7 (Existence of extremisers for S^1 restriction)**.** ***[Proved]*** *The sharp restriction constant $\mathcal{R}$ is attained: there exists $f_* \in L^2(S^1)$ with $\|f_*\|_2 = 1$ and $\|Ef_*\|_{L^6} = \mathcal{R}$.*

Proof. Let f_j be a maximising sequence: $\|f_j\|_2 = 1$, $\Lambda_j := \|Ef_j\|_6^6 \to \mathcal{R}^6 > 0$. Apply the profile decomposition (Theorem 9.3).

Step 1: at least one profile. Since $\left\|Er_j^K\right\|_6 \to 0$ as $K \to \infty$ while $\Lambda_j \to \mathcal{R}^6 > 0$, the decoupling (15) forces $\phi^1 \neq 0$.

Step 2: no splitting. By the restriction inequality, $\left\|E\phi^k\right\|_6^6 \leq \mathcal{R}^6 \left\|\phi^k\right\|_2^6$. Set $\alpha_k := \left\|\phi^k\right\|_2$ and $a_k := \alpha_k^2$. The decoupling gives $\mathcal{R}^6 \leq \mathcal{R}^6 \sum_k a_k^3$. From (14), $\sum_k a_k \leq 1$.

If $M \geq 2$ profiles are nonzero, all $a_k > 0$ and $\sum a_k \leq 1$. But the multinomial expansion $\left(\sum_k a_k\right)^3 = \sum_k a_k^3 + 3\sum_{k\neq l} a_k^2 a_l + 6\sum_{k<l<m} a_k a_l a_m$ shows $\sum_k a_k^3 < \left(\sum_k a_k\right)^3 \leq 1$ (since the cross terms are strictly positive). This gives $\mathcal{R}^6 < \mathcal{R}^6$, a contradiction.

Step 3: convergence. With a single profile: $f_j = g_j^{-1} \cdot \phi^1 + r_j$, $\|r_j\|_2 \to 0$, $\|\phi^1\|_2 = 1$. Hence $g_j \cdot f_j \to \phi^1$ strongly, and ϕ^1 is a maximiser. □

Remark 9.8 (The homogeneity gap). The argument uses $6 > 2$: the L^6 norm is superquadratic in f, making $\sum a_k^3 < (\sum a_k)^3$ when there are multiple profiles. This mechanism is specific to maximising sequences and does not extend to sub-maximal critical points.

9.5 The splitting obstruction for general critical sequences

For a general sequence of critical points with $\Lambda_j \to \Lambda_\infty \in [\varepsilon, \mathcal{R}^6)$, the strict-convexity argument breaks down: multi-profile decompositions are *algebraically consistent* with the Euler–Lagrange equation.

Proposition 9.9 (Structure of multi-profile decompositions at critical sequences)**.** ***[Proved]*** *Suppose a sequence of normalised critical points with $\Lambda_j \to \Lambda_\infty > 0$ admits a profile decomposition with $M \geq 2$ nonzero profiles $\phi^1, \ldots, \phi^M$. Then:*

1. *Each normalised profile $\psi^k = \phi^k/\alpha_k$ (where $\alpha_k := \left\|\phi^k\right\|_2$) is a critical point with eigenvalue $\widetilde{\Lambda}_k = \Lambda_\infty/\alpha_k^4 > \Lambda_\infty$.*
2. *The number of profiles satisfies $M \leq (\mathcal{R}^6/\varepsilon)^{1/2}$.*
3. *The identity $\sum_k \alpha_k^6 \widetilde{\Lambda}_k = \Lambda_\infty \sum_k \alpha_k^2 = \Lambda_\infty$ holds automatically, so the energy decomposition produces no contradiction.*

Proof. (1) Proposition 9.6 gives $E^*(|E\phi^k|^4 E\phi^k) = \Lambda_\infty \phi^k$. Rescaling: $E^*(|E\psi^k|^4 E\psi^k) = (\Lambda_\infty/\alpha_k^4)\psi^k$, so $\widetilde{\Lambda}_k = \Lambda_\infty/\alpha_k^4$. Since $\alpha_k^2 < \sum_l \alpha_l^2 = 1$, we have $\alpha_k < 1$ and $\widetilde{\Lambda}_k > \Lambda_\infty$.

(2) From $\widetilde{\Lambda}_k \leq \mathcal{R}^6$: $\alpha_k^4 \geq \Lambda_\infty/\mathcal{R}^6 \geq \varepsilon/\mathcal{R}^6$, so $\alpha_k^2 \geq (\varepsilon/\mathcal{R}^6)^{1/2}$. The constraint $\sum_k \alpha_k^2 = 1$ gives the bound on M.

(3) $\sum_k \alpha_k^6 \widetilde{\Lambda}_k = \sum_k \alpha_k^6 \cdot \Lambda_\infty/\alpha_k^4 = \Lambda_\infty \sum_k \alpha_k^2 = \Lambda_\infty$. □

Part (3) is the central difficulty: **the Euler–Lagrange equation alone does not prevent a critical sequence from splitting into bubbles at diverging locations, each individually a critical point at a higher eigenvalue level.** This is the open content of Theorem 9.10:

Theorem 9.10 (Compactness modulo G)**.** ***[Open]*** *Let (f_j, Λ_j) be normalised critical points with $\Lambda_j \geq \varepsilon > 0$. Then there exist $g_j \in G$ and a subsequence such that $g_j \cdot f_j \to f_\infty$ strongly in $L^2(S^1)$ and $\Lambda_j \to \Lambda_\infty = \|Ef_\infty\|_6^6$.*

Equivalently: every critical sequence has **exactly one nonzero profile** in its decomposition.

9.6 Approaches to rule out splitting

We describe three general strategies that may close the gap, and then an extremiser-specific refinement using nonnegative representatives.

(A) Descending induction on critical values. At the maximal level $\Lambda = \mathcal{R}^6$, Theorem 9.7 proves single-profile concentration. Suppose inductively that compactness modulo G holds for critical sequences with $\Lambda_j \to \Lambda'$ for all $\Lambda' > \Lambda_\infty$. A splitting at level Λ_∞ produces profiles at levels $\widetilde{\Lambda}_k > \Lambda_\infty$, where the inductive hypothesis already gives finiteness of shapes. The splitting identity $\sum_k \alpha_k^2 = 1$, $\widetilde{\Lambda}_k = \Lambda_\infty/\alpha_k^4$ imposes algebraic constraints relating the profile masses to the (now finitely many) critical values above Λ_∞.

Difficulty: the base case gives one shape (the maximiser); but the inductive step requires both finiteness and nondegeneracy at every intermediate level, creating a dependency between this theorem and the results of Section 8. Breaking this circularity requires either a direct nondegeneracy argument at all levels, or an independent route to single-profile concentration.

(B) Pohozaev–Morawetz monotonicity. A localised identity for $u = c(|u|^4 u * \widehat{\sigma})$ could yield a monotonicity formula controlling the spatial distribution of $|u|^6$ and preventing mass separation. The starting point would be to multiply the equation by a radial weight $a(|x|)\,\bar{u}$ and integrate; the resulting identity relates $\int a'|u|^6$ to boundary terms and the nonlocal convolution structure.

Difficulty: the convolution against $\widehat{\sigma}$ (which decays only like $|x|^{-1/2}$) introduces long-range interactions that resist clean localisation.

(C) Unique continuation for the integrated equation. If one can show that two critical points ϕ^1, ϕ^2 at levels $\widetilde{\Lambda}_1, \widetilde{\Lambda}_2 > \Lambda_\infty$ cannot coexist as profiles of a common critical sequence—i.e., the relation

$$f_j \approx e^{-ix_j^1\cdot\omega}\phi^1 + e^{-ix_j^2\cdot\omega}\phi^2, \qquad |x_j^1 - x_j^2| \to \infty,$$

contradicts $E^*(|Ef_j|^4 Ef_j) = \Lambda_j f_j$ at order beyond the profile-level Euler–Lagrange—this would rule out splitting. The relevant obstruction would arise from cross terms at intermediate spatial scales $|x| \sim |x_j^1 - x_j^2|^{1/2}$ where the two Herglotz waves overlap at polynomial (not exponential) level.

(D) Positive gauge for the extremiser search. For the sharp constant, Lemma 6.3 allows the search to be restricted to nonnegative critical points. This is not available for arbitrary critical sequences, but it removes much of the translation ambiguity: a nonnegative representative cannot be translated nontrivially and remain nonnegative.

Lemma 9.11 (Nonnegativity fixes the translation parameter)**.** *Let* $0 \neq f \in L^2(S^1)$ *with* $f \geq 0$ *a.e. If*

$$e^{i\alpha} e^{-ix_0\cdot\omega} f(R^{-1}\omega) \geq 0 \qquad \text{for a.e. } \omega \in S^1,$$

then $x_0 = 0$*. In that case the remaining phase must satisfy* $e^{i\alpha} = 1$ *on the support of* f*.*

Proof. The set where $f(R^{-1}\omega) > 0$ has positive measure. On this set, the phase $e^{i\alpha}e^{-ix_0\cdot\omega}$ must equal 1. If $x_0 \neq 0$, the set $\{\omega \in S^1 : e^{i\alpha}e^{-ix_0\cdot\omega} = 1\}$ is finite, since it is a finite union of level sets $x_0 \cdot \omega = \alpha + 2\pi m$. This cannot contain a positive-measure subset of S^1. Hence $x_0 = 0$. □

The profile-decomposition consequence is the following. By the gauge-fixed bootstrap in Proposition 9.1, a nonnegative critical sequence with $\Lambda_j \geq \varepsilon$ satisfies $\|f_j\|_{C^1(S^1)} \leq C_\varepsilon$. Thus every escaping profile vanishes: for $|x_j| \to \infty$ and $h \in C^1(S^1)$, stationary phase gives

$$\int_{S^1} e^{ix_j\cdot\omega} f_j(\omega)\overline{h(\omega)}\, d\sigma(\omega) = O_{C_\varepsilon,h}(|x_j|^{-1/2}),$$

and density extends this weak convergence to all $h \in L^2(S^1)$. In the iterative profile construction, after subtracting finitely many previously selected profiles, the terms involving $e^{i(x_j^k - x_j^\ell)\cdot\omega}\phi^\ell$ also vanish by the Riemann–Lebesgue lemma whenever the translation parameters diverge. Thus a candidate profile with $|x_j^k| \to \infty$ has zero weak limit.

Thus the extra input needed for the extremiser search is not a new profile-decomposition mechanism, but the standard gauge-fixed bootstrap: the L^2 normalisation, the equation, and the lower bound on Λ give the uniform norms, while positivity supplies the gauge because pure translation orbits leave the nonnegative slice by Lemma 9.11. This closes the escaping-profile obstruction for the extremiser search without proving compactness of the full critical set.

Remark 9.12 (Sufficiency for identifying $\mathcal{R}$)*.* For the purpose of *computing* the sharp constant $\mathcal{R}$, the full Theorem 9.10 is not needed. Theorem 9.7 gives existence of a maximiser, and if the maximiser is unique modulo G (e.g., the constant function $f_0 = (2\pi)^{-1/2}$), then $\mathcal{R} = \|Ef_0\|_{L^6}$ and the problem is solved. The open compactness theorem would additionally yield finiteness of *all* critical values above ε, which is a stronger structural result.

Remark 9.13 (Accumulation of critical values at zero)*.* Whether critical values can accumulate at 0 is unknown. The compactness + nondegeneracy programme gives finiteness only above a fixed threshold. For the sharp constant, it suffices to work above any ε smaller than the maximum critical value.

10 From the Three Inputs to the Sharp Constant

10.1 Finiteness after the positivity reduction

Theorem 10.1 (Finiteness of positive critical values in compact windows)**.** ***[Proved]*** *Fix $0 < \Lambda_- < \Lambda_+ < \infty$. Then the set*

$$\mathcal{V}_+(\Lambda_-, \Lambda_+) := \left\{ \Lambda \in [\Lambda_-, \Lambda_+] : \begin{array}{l} \exists f \in L^2(S^1),\ f \geq 0,\ \|f\|_2 = 1, \\ E^*(|Ef|^4 Ef) = \Lambda f \end{array} \right\}$$

is finite.

Proof. Let

$$\mathcal{C}_+ := \left\{ (f, \Lambda) : f \geq 0,\ \|f\|_2 = 1,\ \Lambda \in [\Lambda_-, \Lambda_+],\ E^*(|Ef|^4 Ef) = \Lambda f \right\}.$$

Step 1: compactness of the positive slice. By the gauge-fixed bootstrap, Proposition 9.1, for every $m \geq 0$ there is a constant $C_m = C_m(\Lambda_-)$ such that $\|f\|_{C^m(S^1)} \leq C_m$ for every $(f, \Lambda) \in \mathcal{C}_+$. The reason this applies without an additional translation choice is precisely Lemma 9.11: the nonnegative slice fixes the noncompact translation parameter.

Given a sequence $(f_j, \Lambda_j) \in \mathcal{C}_+$, pass to a subsequence with $\Lambda_j \to \Lambda_\infty \in [\Lambda_-, \Lambda_+]$. The uniform C^{m+1} bound and Arzelà–Ascoli give, after passing to a further subsequence, $f_j \to f_\infty$ in $C^m(S^1)$ for each fixed m (and in particular in L^2). Then $f_\infty \geq 0$ and $\|f_\infty\|_2 = 1$. Since

$$T(f) := E^*(|Ef|^4 Ef)$$

is continuous $L^2(S^1) \to L^2(S^1)$ by Stein–Tomas and Hölder, passing to the limit in $T(f_j) = \Lambda_j f_j$ gives $T(f_\infty) = \Lambda_\infty f_\infty$. Hence $(f_\infty, \Lambda_\infty) \in \mathcal{C}_+$. Thus $\mathcal{C}_+$ is compact in $L^2(S^1) \times \mathbb{R}$.

Step 2: local finite-dimensional analytic structure. Work on the real Hilbert space $H := L^2(S^1; \mathbb{C})$ with real inner product $\operatorname{Re}\langle \cdot, \cdot \rangle$. Define

$$\Phi : H \times \mathbb{R} \to H \times \mathbb{R}, \qquad \Phi(f, \Lambda) := \big(T(f) - \Lambda f,\ \|f\|_2^2 - 1\big).$$

The map T is a real-analytic quintic polynomial map: $T(f) = E^*((Ef)^3(\overline{Ef})^2)$. At a zero (f_0, Λ_0),

$$D\Phi_{(f_0,\Lambda_0)}(h, \mu) = ((K_{u_0} - \Lambda_0 I)h - \mu f_0,\ 2\operatorname{Re}\langle f_0, h \rangle), \qquad u_0 = Ef_0.$$

By Corollary 7.4, $K_{u_0} - \Lambda_0 I$ is Fredholm of index 0. The displayed derivative is a finite-rank perturbation of $(h, \mu) \mapsto ((K_{u_0} - \Lambda_0 I)h, \mu)$, and is therefore Fredholm of index 0.

The analytic Fredholm Lyapunov–Schmidt reduction now applies: near each zero of Φ, the zero set $\Phi^{-1}(0)$ is homeomorphic to the zero set of a finite-dimensional

real-analytic map. By the standard local structure theorem for real-analytic sets (equivalently, Lojasiewicz local finiteness), this local zero set has only finitely many connected components, and each such component is locally arc-connected through finitely many analytic arcs.

Step 3: the value is constant on local critical components. Let $J(f) := \|Ef\|_{L^6}^6$. On $\Phi^{-1}(0)$ one has $\Lambda = J(f)$, because

$$\Lambda \|f\|_2^2 = \operatorname{Re}\langle T(f),\, f\rangle = \int_{\mathbb{R}^2} |Ef(x)|^6\, dx.$$

If $t \mapsto (f(t), \Lambda(t))$ is a C^1 arc in $\Phi^{-1}(0)$, then $\|f(t)\|_2 = 1$ and $T(f(t)) = \Lambda(t)f(t)$, so

$$\frac{d}{dt} J(f(t)) = 6 \operatorname{Re}\langle T(f(t)),\, f'(t)\rangle = 6\Lambda(t) \operatorname{Re}\langle f(t),\, f'(t)\rangle = 0.$$

Therefore $J(f(t))$, and hence $\Lambda(t)$, is constant on every such arc. Since the local analytic components are connected by finite chains of these arcs, Λ is constant on each local connected component of $\Phi^{-1}(0)$.

Finally, compactness of $\mathcal{C}_+$ gives a finite cover by neighbourhoods of the type described in Step 2. Each neighbourhood contains only finitely many local critical components, and Step 3 gives at most one value of Λ on each of them. Hence the projection of $\mathcal{C}_+$ onto the Λ coordinate, namely $\mathcal{V}_+(\Lambda_-, \Lambda_+)$, is finite. □

10.2 A spectral gap around the constant value

Let

$$f_0 = (2\pi)^{-1/2}, \qquad \Lambda_0 := \|Ef_0\|_{L^6}^6 = 16\pi^4 \int_0^\infty J_0(r)^6\, r\, dr.$$

If C_{ST} denotes any explicit Stein–Tomas bound for $E : L^2(S^1) \to L^6(\mathbb{R}^2)$, then every normalised critical value satisfies $\Lambda \le C_{\mathrm{ST}}^6$. Thus, when working with the eigenvalue parameter Λ rather than with the restriction norm $\Lambda^{1/6}$, the natural finite window for the extremiser search is

$$W_0 := \left[\frac{1}{2}\Lambda_0,\ C_{\mathrm{ST}}^6\right].$$

By Theorem 10.1, $\mathcal{V}_+(W_0) := \mathcal{V}_+(\Lambda_0/2, C_{\mathrm{ST}}^6)$ is a finite set and contains Λ_0. Hence the nonlinear spectral gap around the constant value,

$$\gamma_0 := \operatorname{dist}\big(\Lambda_0,\, \mathcal{V}_+(W_0) \setminus \{\Lambda_0\}\big), \tag{18}$$

is strictly positive, with the convention $\gamma_0 = +\infty$ if $\mathcal{V}_+(W_0) = \{\Lambda_0\}$.

The proof of Theorem 10.1 gives positivity of γ_0, but not a numerical lower bound. To make the gap quantitative, one needs to quantify two pieces.

Local isolation at the constant. On the slice orthogonal to phase and translations, define the linear nondegeneracy gap

$$\nu_0 := \min\left\{\frac{4}{5}\Lambda_0,\ \inf_{n\geq 2}|\mathcal{I}_n - \Lambda_0|\,,\ \inf_{n\geq 2}|5\mathcal{I}_n - \Lambda_0|\right\}, \qquad \mathcal{I}_n = 16\pi^4\mathcal{J}_n.$$

If interval arithmetic proves $\nu_0 \geq \underline{\nu} > 0$ and $M_0(r)$ is a Lipschitz constant for the derivative of the sliced Euler–Lagrange map in the L^2 ball $\|f - f_0\|_2 \leq r$, then the Newton–Kantorovich estimate

$$r_0 \leq \frac{\underline{\nu}}{2M_0(r_0)}$$

certifies that the constant is the only positive critical point in that ball, up to the trivial rotation symmetry of the constant.

A rough numerical quadrature of the Bessel integrals gives

$$\int_0^\infty J_0(r)^6 r\,dr \approx 0.33678766, \qquad \Lambda_0 \approx 524.899, \qquad \Lambda_0^{1/6} \approx 2.84018,$$

and suggests

$$\frac{\nu_0}{\Lambda_0} = \min_{n\geq 2}\left\{\left|\frac{\mathcal{I}_n}{\Lambda_0} - 1\right|, \left|\frac{5\mathcal{I}_n}{\Lambda_0} - 1\right|\right\} \approx 0.452,$$

with the minimum apparently coming from the $5\mathcal{I}_2 - \Lambda_0$ branch. A reasonable first rigorous target is therefore

$$\nu_0 \geq 0.4\,\Lambda_0. \tag{19}$$

This target only requires a finite interval-arithmetic check for the small modes, together with the already proved tail estimate $\mathcal{J}_n \to 0$ for large n.

Current certified finite-mode check. The script `numerics/circle_gap_flint_verify.py` gives a first ball-arithmetic certificate for the small-mode part of this target. It uses `python-flint` Arb balls, evaluates the Bessel functions at the midpoints of the radial mesh, and encloses the variation on each interval by the elementary bounds $|J_m| \leq 1$ and $|J_m'| \leq 1$ for integer m on the real line. With cutoff $R = 200$ and the conditional tail majorant $|J_m(r)| \leq (2/(\pi r))^{1/2}$ on $[R, \infty)$, the current runs prove

$$\frac{\mathcal{J}_2}{\mathcal{J}_0} \leq 0.117364, \qquad \frac{\mathcal{J}_3}{\mathcal{J}_0} \leq 0.091870, \qquad \frac{\mathcal{J}_n}{\mathcal{J}_0} \leq 0.072366 \quad (4 \leq n \leq 12).$$

The first number is the bottleneck and was obtained with mesh $h = 0.0025$; the other displayed bounds use $h = 0.01$. In particular, for $2 \leq n \leq 12$ the finite-mode certificate proves

$$\frac{\mathcal{J}_n}{\mathcal{J}_0} < 0.12,$$

and hence

$$|5\mathcal{I}_n - \Lambda_0| > 0.4\Lambda_0, \qquad |\mathcal{I}_n - \Lambda_0| > 0.4\Lambda_0 \qquad (2 \leq n \leq 12).$$

What remains for a complete proof of (19) is to replace or quote the tail majorant in a final form and prove the large-mode estimate for all $n \geq 13$.

Global separation from the rest of the positive window. After the local ball $\|f - f_0\|_2 < r_0$ has been certified, the remaining set

$$\mathcal{C}_+^{\text{far}} := \left\{ (f, \Lambda) : \begin{array}{l} f \geq 0,\ \|f\|_2 = 1,\ E^*(|Ef|^4 Ef) = \Lambda f, \\ \Lambda \in W_0,\ \|f - f_0\|_2 \geq r_0 \end{array} \right\}$$

is compact. A validated search on this compact set should return finitely many disjoint value intervals $B_1, \ldots, B_N$ containing every positive critical value in the far region. Then the explicit certificate

$$\gamma_{\text{cert}} := \min_{1 \leq j \leq N,\ \Lambda_0 \notin B_j} \operatorname{dist}(\Lambda_0, B_j)$$

is a rigorous lower bound for γ_0.

Thus the current proof gives the qualitative gap (18). The practical quantitative route is: certify the local linear target (19), use it to remove a ball around the constant by Newton–Kantorovich, and then use the interval search on the compact far region to produce γ_{cert}.

10.3 What a rigorous brute-force search would mean

The phrase "search the compact region" should not be interpreted as requiring a prior description of the compact set. The algorithm searches a computable *outer enclosure* of that compact set. The compactness theorem supplies the analytic reason such an enclosure can be made finite; the actual numerical proof must supply explicit constants.

There are two complementary brute-force targets.

Eigenvalue exclusion. Fix an interval $I \subset W_0$ for Λ. A rigorous search attempts to prove:

$$\nexists f \geq 0, \quad \|f\|_2 = 1, \quad E^*(|Ef|^4 Ef) = \Lambda f, \quad \Lambda \in I,$$

except possibly inside boxes already validated as containing known solutions. The output of such a search is not "success/failure" in the usual numerical sense. Each box in the finite cover is labelled *empty*, *validated*, or *undecided*. A theorem is obtained only if no undecided boxes remain.

Direct upper-bound search. Instead of solving the Euler–Lagrange equation, one can try to prove the stronger variational statement

$$\sup_{\substack{f \geq 0 \\ \|f\|_2 = 1}} \|Ef\|_{L^6}^6 \leq \Lambda_0 + \eta.$$

If an eigenvalue gap $\gamma_0 > \eta$ has already been certified in the positive window, then this upper bound forces the maximiser to have value Λ_0. This route may be computationally easier because it asks for upper bounds on a scalar polynomial functional, rather than for a complete classification of roots of a vector equation.

Fourier truncation and tails. Write

$$f(\theta) = \sum_{n\in\mathbb{Z}} a_n e^{in\theta}, \qquad P_N f = \sum_{|n|\le N} a_n e^{in\theta}, \qquad r_N = f - P_N f.$$

For real-valued nonnegative candidates, $a_{-n} = \overline{a_n}$; if one also imposes an even symmetry ansatz, the unknowns reduce further to real cosine coefficients. The bootstrap gives an explicit tail estimate of the form

$$\|r_N\|_{L^2} \le \tau_N(B_m), \qquad \tau_N(B_m) \to 0,$$

where B_m is a certified C^m or Sobolev bound for positive critical points in the window. Thus one searches over the finite coefficient vector $(a_{-N}, \dots, a_N)$ and an interval uncertainty representing r_N.

The finite-dimensional residual is

$$F_k(a, \Lambda) := \widehat{T(P_N f)}(k) - \Lambda a_k, \qquad |k| \le N,$$

with an explicit interval error for all terms containing at least one factor of r_N. A convenient crude tail bound follows from Stein–Tomas and the Lipschitz estimate for $z \mapsto |z|^4 z$:

$$\|T(f) - T(P_N f)\|_2 \le C\, C_{\mathrm{ST}}^6 \big(\|f\|_2 + \|P_N f\|_2\big)^4 \|r_N\|_2 ,$$

with an absolute constant C. This is probably not sharp enough for the final proof, but it is the correct first enclosure. The Bessel-product technology should be used later to reduce this overestimate.

Why sixfold Bessel products appear. The extension of one Fourier mode is

$$E(e^{in\theta})(r, \phi) = 2\pi i^{|n|} J_{|n|}(r) e^{in\phi}.$$

Consequently, the coefficient $\widehat{T(P_N f)}(k)$ is a finite quintic sum over indices satisfying the angular selection rule

$$n_1 + n_2 + n_3 - n_4 - n_5 - k = 0,$$

with constants given by radial integrals of six Bessel functions:

$$\int_0^\infty J_{|n_1|}(r) J_{|n_2|}(r) J_{|n_3|}(r) J_{|n_4|}(r) J_{|n_5|}(r) J_{|k|}(r)\, r\, dr.$$

These constants must be enclosed rigorously. This is exactly the type of input studied by Oliveira e Silva–Thiele in their estimates for sixfold Bessel products, and by the band-limited maximiser papers described below.

Interval branch-and-bound. The finite search region is a box in

$$(a_{-N}, \ldots, a_N, \Lambda),$$

augmented by interval constraints for normalisation, positivity, and the tail. Positivity can be enforced by interval evaluation on arcs: if $P_N f$ is bounded below by more than the certified tail and mesh interpolation error, the box is positive; if it is forced negative somewhere, the box is discarded; otherwise it is subdivided.

For the eigenvalue search, each box is processed as follows.

1. Evaluate the interval residuals F_k and the normalisation residual. If some residual interval excludes 0, discard the box.

2. Apply an interval Newton or Krawczyk operator to the finite system. If it proves no zero lies in the box, discard it. If it maps strictly into the box, validate a unique zero.

3. If neither test succeeds, subdivide the box, unless the box is already below the target resolution; in that case it is marked undecided and no theorem is claimed.

For the variational upper-bound search, the analogous step is interval upper-bounding of the scalar functional $\|E(P_N f)\|_6^6$ plus the tail error. Boxes whose upper bound is below $\Lambda_0 + \eta$ are discarded; boxes near the top are subdivided or handled by local concavity at the constant.

How this could prove the sharp constant. A plausible staged proof would be:

1. Certify the local linear gap target $\nu_0 \geq 0.4\Lambda_0$ by interval Bessel integration.

2. Use Newton–Kantorovich to remove an explicit ball around f_0.

3. Run the eigenvalue exclusion search on $[\Lambda_0, \Lambda_0 + \gamma]$ outside that ball. If it exhausts the boxes, then no positive critical value lies in this interval.

4. Separately prove by branch-and-bound that $\|Ef\|_6^6 \leq \Lambda_0 + \gamma/2$ for all positive normalised candidates in the compact window.

5. Since an extremiser exists and may be chosen nonnegative, the maximising value must then equal Λ_0.

The same architecture can be run in restriction-norm units $\Lambda^{1/6}$. One should be careful about scales: near Λ_0, a gap of 10^{-3} in $\Lambda^{1/6}$ corresponds to a gap of order $6\Lambda_0^{5/6} \cdot 10^{-3}$ in Λ.

Relevant numerical and analytic references. The closest existing works are:

- D. Oliveira e Silva and C. Thiele, *Estimates for certain integrals of products of six Bessel functions*, Rev. Mat. Iberoam. 33 (2017), no. 4, 1423–1462, doi:10.4171/RMI/977. This is the main reference for rigorous estimates of the sixfold Bessel integrals needed in the residual coefficients.

- E. Carneiro, D. Foschi, D. Oliveira e Silva and C. Thiele, *A sharp trilinear inequality related to Fourier restriction on the circle*, Rev. Mat. Iberoam. 33 (2017), no. 4, 1463–1486, doi:10.4171/RMI/978. This uses the sixfold Bessel estimates and proves, among other things, that constants are local extremizers for the circle Tomas–Stein problem.

- D. Oliveira e Silva, C. Thiele and P. Zorin-Kranich, *Band-limited maximizers for a Fourier extension inequality on the circle*, Exp. Math. 31 (2022), no. 1, 192–198, doi:10.1080/10586458.2019.1596847. They prove uniqueness of constants among real-valued functions with Fourier modes up to degree 30.

- J. Barker, C. Thiele and P. Zorin-Kranich, *Band-limited maximizers for a Fourier extension inequality on the circle, II*, Exp. Math. 32 (2023), no. 2, 280–293, doi:10.1080/10586458.2021.1926011. This pushes the band-limited computation to modes up to degree 120.

- R. E. Moore, R. B. Kearfott and M. J. Cloud, *Introduction to Interval Analysis*, SIAM, 2009. This is a standard reference for interval Newton and Krawczyk methods.

- F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. 66 (2017), no. 8, 1281–1292, doi:10.1109/TC.2017.2690633. Arb/ball arithmetic is a natural implementation platform for rigorous Bessel-integral and polynomial residual enclosures.

Theorem 10.2 (Conditional finiteness of shapes)**.** ***[Conditional]*** *Assume:*

1. *Compactness modulo G (Theorem 9.10) holds for critical points with $\Lambda \geq \varepsilon$;*

2. *Every such critical point is nondegenerate modulo G (i.e.,* $\ker L = T_{f_0}(G \cdot f_0)$ *on the tangent space to the unit sphere).*

Then the set of shapes $\{[f, \Lambda] \in \mathcal{Z} : \Lambda \geq \varepsilon\}$ is finite. In particular, the set of critical values above ε is finite.

Proof. By compactness modulo G, the quotient $\mathcal{Z}_\varepsilon := \{f : \text{critical}, \Lambda \geq \varepsilon\}/G$ is compact.

By nondegeneracy: at each $[f_0] \in \mathcal{Z}_\varepsilon$, choose a slice S_{f_0} complementary to $T_{f_0}(G \cdot f_0)$ in the tangent space to $\|f\|_2 = 1$. The condition $\ker L|_{S_{f_0}} = \{0\}$ plus Fredholm index

zero implies $L|_{S_{f_0}}$ is invertible with bounded inverse $\left\| L^{-1}|_{S_{f_0}} \right\| \leq \mu_{f_0}^{-1}$. By the implicit function theorem, $[f_0]$ is an isolated point of $\mathcal{Z}_\varepsilon$.

A compact set all of whose points are isolated is finite. □

Corollary 10.3 (Identification of the sharp constant)**.** *Under the hypotheses of Theorem 10.2, the sharp restriction constant is*

$$\mathcal{R} = \max\{\Lambda_j^{1/6} : \Lambda_j \geq \varepsilon\}$$

for any ε smaller than all critical values in the finite list. The maximiser is the shape achieving the largest Λ_j.

10.4 What remains for the sharp constant on S^1

Two paths are available, depending on the desired strength of the result.

Path 1: Identify $\mathcal{R}$ (the sharp constant).

1. **Verify Bessel integrals.** Check $\mathcal{J}_n \neq \mathcal{J}_0$ and $5\mathcal{J}_n \neq \mathcal{J}_0$ for $n = 2, 3, \ldots, n_*$ by interval arithmetic. This establishes nondegeneracy at the constant solution. *Routine numerical computation.*

2. **Restrict the search to nonnegative eigenfunctions.** By Lemma 6.3, any extremiser can be replaced by a nonnegative extremiser, possibly in a different shape. Therefore, for the sharp constant, one only needs to retain those candidate values Λ_j for which the nonlinear eigenvalue equation has at least one nonnegative solution. The positive gauge also removes exact translation ambiguity by Lemma 9.11, and Theorem 10.1 proves that the retained critical values in any fixed positive window are finite.

3. **Show the constant is the unique maximiser.** Theorem 9.7 gives existence of a maximiser. If one can prove that the constant function $f_0 = (2\pi)^{-1/2}$ is the unique maximiser modulo G—e.g., by a rearrangement or symmetrisation argument, or by ruling out non-radial maximisers via the eigenvalue analysis—then $\mathcal{R} = \|Ef_0\|_{L^6}$ and the programme is complete. *This bypasses compactness of the full critical set.*

Path 2: Finiteness of all critical values above ε (the full structural result).

1. Bessel integrals (same as above).

2. **Rule out multi-profile splitting (Theorem 9.10).** The profile decomposition is in place (Theorem 9.3), and each profile inherits the Euler–Lagrange equation (Proposition 9.6). The open step is to show that no critical sequence admits

more than one profile. Section 9 identifies three general strategies: descending induction, Pohozaev–Morawetz monotonicity, and unique continuation for cross terms. *Requires new analytic input.*

3. **Enumerate non-constant shapes (if any).** Either prove that the constant is the only critical shape, or verify nondegeneracy at all other shapes.

Part III
Extensions and Supplementary Material

11 The Higher-Dimensional Transversality Route

For $n \geq 2$ and the equation $\rho = cT(\rho)$ on $L^2(S^{n-1})$, Proposition 4.1 forces $\widehat{u}$ onto S^{n-1}, and the problem reduces to the incidence-fibre equation (3). The symmetry group is $G = \mathbb{R}^n \rtimes SO(n)$, acting on densities by $(x_0, R) \cdot \rho(\omega) = e^{-ix_0\cdot\omega}\rho(R^{-1}\omega)$.

11.1 The five-step programme

The route to finiteness of shapes requires five analytic inputs. We state each and assess its status for general n.

Step 1: Smoothing / Fredholm estimate. **[Open]** The target is a multilinear smoothing estimate

$$A_k : H^s(S^{n-1})^k \to H^{s+\varepsilon}(S^{n-1}), \qquad \|A_k(\rho_1, \ldots, \rho_k)\|_{H^{s+\varepsilon}} \leq C_s \prod_{j=1}^{k} \|\rho_j\|_{H^s},$$

for some $\varepsilon > 0$, where A_k is the k-linear operator behind the incidence-fibre equation. A weaker but sufficient target is compactness of A_k on bounded sets.

For the S^1 problem, the truncation argument (Theorem 7.3) bypasses this step entirely. For $n \geq 2$, the proof must exploit the curvature of S^{n-1} and the transversality of the constraint $\omega_1 + \cdots + \omega_k = \omega$. The key obstruction is the existence of *degenerate fibres*; see Section 11.2.

Step 2: Regularity bootstrap. **[Open]** Show that weak L^2 solutions are automatically smooth: $\rho \in L^2(S^{n-1}),\ \rho = cT(\rho) \implies \rho \in H^s$ for all s. If Step 1 gives $T : L^2 \to H^\varepsilon$, iteration yields the bootstrap.

Step 3: Local slices for the symmetry action. [Open] At a solution ρ, construct a complement S_ρ to the orbit tangent

$$\mathfrak{g} \cdot \rho = \{-i(a \cdot \omega)\rho - (A\omega) \cdot \nabla_{S^{n-1}}\rho : a \in \mathbb{R}^n,\ A \in \mathfrak{so}(n)\}.$$

This must be done separately on orbit-type strata (radial, zonal, fully asymmetric solutions have different stabiliser dimensions).

Step 4: Nondegeneracy modulo symmetry. [Open] Verify $\ker L_\rho = \mathfrak{g} \cdot \rho$ at each solution, where $L_\rho = I - ckA_k(\rho, \ldots, \rho, \cdot)$. For the S^1 problem at the constant solution, this reduces to Bessel integrals (Theorem 8.4).

Step 5: Compactness modulo G. [Open] Prove sequential compactness of the solution set modulo G:

$$\rho_j = cT(\rho_j) \implies \exists g_j \in G :\ g_j \cdot \rho_j \to \rho_\infty.$$

This requires a priori bounds ($\|\rho\| \le R$), a non-collapse lower bound ($\|\rho\| \ge r > 0$), and a no-escape condition for translations. Rotations are compact; translations are the genuine noncompact obstruction.

11.2 Degenerate fibres

The fibre $V_k(\omega) = \{(\omega_1, \ldots, \omega_k) \in (S^{n-1})^k : \sum \omega_j = \omega\}$ is generically a smooth manifold of dimension $(n-1)k - (n-1+1) = (n-1)(k-1) - 1$... when the differential of $\Sigma : \omega_1 + \cdots + \omega_k$ has full rank. Degeneracies occur when all tangent lines $T_{\omega_j} S^{n-1}$ are parallel.

Example for $k = 5$ on S^1: The configuration $(\omega_0, \omega_0, \omega_0, -\omega_0, -\omega_0)$ with output $\omega = \omega_0$ is degenerate. Here $d\Sigma = 3(\text{tangent at } \omega_0) - 2(\text{tangent at } -\omega_0)$ has rank 1 instead of 2 (the tangents at ω_0 and $-\omega_0$ are parallel on S^1). This is a codimension-1 degeneracy, producing an integrable singularity $\sim |t|^{-1/2}$ in the fibre density.

In higher dimensions, the degenerate locus can have higher codimension (tangents at ω_0 and $-\omega_0$ are parallel in $\mathbb{R}^n$ only when $n = 1$; for $n \ge 2$, anti-parallel tangent *spaces* are generically transverse). This suggests that the smoothing estimate may be easier for $n \ge 3$ than for $n = 2$.

11.3 Orbit dimension and radial rigidity

Proposition 11.1 (Orbit dimension, $n = 3$)**.** *For $G = \mathbb{R}^3 \rtimes SO(3)$ ($\dim G = 6$), the orbit dimension of a solution u is $6 - \dim \mathrm{Stab}_G(u)$. Specifically:*

- *Radial: $d = 3$ (only translations).*
- *Zonal (one axis of symmetry): $d = 5$.*

- *Fully asymmetric:* $d = 6$.

Conjecture 11.2 (Radial rigidity)**.** *For the equation* $u = c(|u|^{2(k-1)}u * \widehat{\sigma})$ *on* $\mathbb{R}^3$*, every non-trivial solution is rotationally invariant about some point (its "centre of mass"). If true, all solutions belong to* 3*-dimensional translation orbits, and the shape count reduces to counting radial profiles.*

Warning 11.3. Conjecture 11.2 is currently *unproved*. The preceding sections on the $n = 3$ problem treated it as established; it should be regarded as a useful heuristic until proved.

12 Supplementary: Gegenbauer Expansion and Galerkin Models

12.1 Gegenbauer expansion

[Heuristic]

For completeness, we record the formal harmonic decomposition. Let $\rho(\omega) = \sum_\ell a_\ell C_\ell^\lambda(\omega \cdot \eta)$ with $\lambda = (n-2)/2$. The product of Gegenbauer polynomials linearises as $C_{\ell_1}^\lambda C_{\ell_2}^\lambda = \sum_j \gamma_j(\ell_1, \ell_2) C_j^\lambda$. If T were a spherical convolution, the Funk–Hecke theorem would give $C_\ell^\lambda *_S \sigma = \lambda_\ell(\sigma) C_\ell^\lambda$. Substituting into $\rho = cT(\rho)$ and equating coefficients would give an algebraic system for $\{a_\ell\}$.

However, as noted in Warning 4.3, the incidence-fibre operator is *not* spherical convolution. The Gegenbauer expansion gives a useful *model* but does not rigorously reduce to finitely many coefficients without separate justification.

12.2 Galerkin approximation

Let $H_L = \operatorname{span}\{Y_{\ell m} : 0 \le \ell \le L\}$ and P_L the $L^2(S^{n-1})$ projection. The projected equation $P_L F(\rho_L) = 0$ is a polynomial system of degree k in $N(L) = (L+1)^2$ unknowns (for $n = 3$).

Proposition 12.1 (Galerkin shape count)**.** *After imposing algebraic slice conditions, the number of Galerkin shapes at truncation* L *is bounded by the BKK mixed volume of the sliced Newton polytopes, in particular by* $k^{N(L)}$ *(Bézout).*

Warning 12.2 (Limit caveat). This is a statement about the fixed-L model. It does not imply finiteness of true shapes as $L \to \infty$ without uniform a priori bounds, no-escape compactness, and stability of roots under high-mode perturbation.

Remark 12.3 (How Galerkin helps). A Galerkin computation becomes rigorous if paired with a high-mode bound $\|(I - P_L)T(\rho)\|_X \le \varepsilon_L \Phi(\|\rho\|_X)$ with $\varepsilon_L \to 0$, plus an inverse bound on the sliced Jacobian at each computed root. A Newton–Kantorovich argument would then validate nearby true solutions and certify completeness.

13 Supplementary: Validated Search Strategy

Assuming the transversality route is completed (modulo G the solution set is finite), the next goal is to *find every shape* by a finite, checkable search.

13.1 Constants needed

1. **A priori radius:** $\|\rho\|_X \leq R(c,k)$.
2. **Non-collapse:** $\|\rho\|_X \geq r(c,k) > 0$.
3. **Regularity/mesh modulus:** $\|\rho\|_{C^m} \leq B_m$.
4. **Validated quadrature:** error $q_h(B_m) \to 0$ for the fibre integral on a mesh of size h.
5. **Slice conditioning:** $\|L_\rho \eta\|_X \geq \mu(\rho)\,\|\eta\|_X$ on S_ρ, with $\mu_* := \inf_\rho \mu(\rho) > 0$.
6. **Lipschitz:** $\|DF(\rho) - DF(\tilde{\rho})\| \leq M\,\|\rho - \tilde{\rho}\|_X$.

13.2 Separation and packing

From the conditioning bound, the quantitative isolation scale is $\delta_* \simeq \mu_*/M$: two distinct shapes cannot lie within δ_* of each other in the slice. A crude packing bound is

$$\#\{\text{shapes}\} \leq \left(1 + \frac{2R_h}{\delta_* - 2\eta_h - 2q_h}\right)^{N_{\text{mesh}}(h)},$$

provided $\delta_* > 2\eta_h + 2q_h$.

13.3 The search pipeline

1. Choose a quasi-uniform mesh on S^{n-1} (or directly on S^1 for the circle problem) and fix gauge conditions.
2. Represent candidates by interval boxes for nodal values plus global regularity bounds.
3. For each box, enclose the residual $F(\rho)$ at mesh points using validated quadrature.
4. Discard boxes with nonzero residual enclosure.
5. Apply Krawczyk/interval Newton to surviving boxes.
6. Validate uniqueness and lift to the full space.
7. Certify completeness: every box discarded or assigned.

13.4 Main difficulties

1. **Translation gauge:** $e^{-ix\cdot\omega}\rho(\omega)$ creates oscillation as $|x| \to \infty$. A canonical centre must be fixed.
2. **Uniform conditioning:** $\mu_* > 0$ may be the hardest constant to obtain without knowing all solutions.
3. **Mesh consistency:** nodal errors must not hide solutions between mesh points.
4. **Dimension:** the effective search dimension grows with mesh size. Symmetry decomposition and adaptive refinement are essential.

13.5 Specialisation to S^1

For the circle problem, the validated search simplifies considerably:

- The mesh is a grid on $[0, 2\pi)$ with N equally spaced points.
- The gauge fixes phase, rotation of S^1, and one translation.
- The fibre integral reduces to a trigonometric integral.
- The Jacobian involves the linearised operator L, which decomposes into Fourier modes for radial u_0.

The direct certified programme for the sharp constant is:

1. Prove a priori bounds: $\|f\|_{L^2} = 1$, $\|f\|_{C^m} \le B_m$, $0 < \Lambda_{\min} \le \Lambda \le C_{\mathrm{ST}}^6$.
2. Use Lemma 6.3 to restrict the extremiser search to boxes containing a nonnegative representative.
3. Fix gauges (phase, rotation, translation); in the nonnegative slice, Lemma 9.11 fixes the translation parameter.
4. Mesh S^1; represent f by interval boxes plus derivative bounds.
5. For each (f, Λ) box, enclose the residual $E^*(|Ef|^4 Ef)(\omega_i) - \Lambda f(\omega_i)$.
6. Discard or validate via interval Newton.
7. Sort validated Λ_j intervals; certify the ordering.

14 Summary: The Logical Dependency Graph

We summarise the dependencies and status of the full programme.

Proved:

- Spectral support forcing ($\widehat{u} \subseteq S^1$)
- 1D case complete
- Compactness of linearised operator K_{u_0} on $L^2(S^1)$
- Fredholm index zero for $L = K - \Lambda I$
- Eigenvalue decomposition at the constant solution
- Nondegeneracy $\Leftrightarrow$ Bessel integrals (11)
- $\mathcal{J}_n \to 0$ (finitely many integrals to check)
- Regularity bootstrap for critical points
- Inverse Strichartz inequality for S^1 extension
- Profiles of critical sequences are critical points
- Existence of a maximiser (Theorem 9.7)
- Positivity reduction for extremisers (Lemma 6.3)
- Finiteness of nonnegative critical values in compact positive windows (Theorem 10.1)

Next concrete steps:

- Compute $\mathcal{J}_n$ for $n = 2, 3, \ldots, n_*$ by interval arithmetic; verify (11)
- In the extremiser search, work with the finite positive-window list from Theorem 10.1
- Quantify the gap around Λ_0 by certifying $\nu_0 \geq 0.4\Lambda_0$ and running the far-region interval search
- Prove the constant is the unique maximiser modulo G (this alone gives $\mathcal{R}$)

Open (harder):

- Rule out multi-profile splitting for non-maximising critical sequences (Theorem 9.10)
- Enumeration / uniqueness of non-constant shapes

Conditional conclusion:

- Compactness + nondegeneracy $\Longrightarrow$ finitely many critical values Λ_j
- $\mathcal{R} = \max_j \Lambda_j^{1/6}$

*Notes compiled from working discussions. Statements marked **[Proved]** carry complete proofs; statements marked **[Conditional]** depend on explicitly stated hypotheses; statements marked **[Open]** require new analytic input.*